

NOTES

DEFINITION : If $L\{f(t)\} = F(s)$ then $f(t)$ is known as the Inverse Laplace Transform or Inverse Transform or simply Inverse of $F(s)$ and is denoted by, $L^{-1}\{F(s)\}$.

- ❖ Therefore, $f(t) = L^{-1}\{F(s)\}$. Here L^{-1} is known as the Inverse Laplace Transform operator and it satisfies $LL^{-1} = L^{-1}L = 1$.
- ❖ Note that the Inverse Laplace Transform of $F(s)$ need not exist for all $F(s)$.

Sl No.	$F(S)$	$f(t)$	Example
1.	$L^{-1}\left[\frac{a}{s}\right]$	a	$L^{-1}\left[\frac{5}{s}\right] = 5$
2.	$L^{-1}\left[\frac{1}{s^n}\right]$	$\frac{t^{n-1}}{(n-1)!}$, if n is a positive integer	$L^{-1}\left[\frac{1}{s^4}\right] = \frac{t^3}{3!}$
		$\frac{t^{n-1}}{\Gamma(n)}$	$L^{-1}\left[\frac{1}{s^{7/2}}\right] = \frac{t^{5/2}}{\Gamma(7/2)}$
3.	$L^{-1}\left[\frac{1}{s-a}\right]$	e^{at}	$L^{-1}\left[\frac{1}{s-8}\right] = e^{8t}$
4.	$L^{-1}\left[\frac{1}{s+a}\right]$	e^{-at}	$L^{-1}\left[\frac{1}{s+1}\right] = e^{-t}$
6.	$L^{-1}\left[\frac{s}{s^2+a^2}\right]$	$\cos at$	$L^{-1}\left[\frac{s}{s^2+25}\right] = \cos 5t$
7.	$L^{-1}\left[\frac{1}{s^2+a^2}\right]$	$\frac{1}{a} \sin at$	$L^{-1}\left[\frac{1}{s^2+16}\right] = \frac{1}{4} \sin 4t$

8.	$L^{-1}\left[\frac{s}{s^2 - a^2}\right]$	<i>cos hat</i>	$L^{-1}\left[\frac{s}{s^2 - 36}\right] = \cosh 6t$
9.	$L^{-1}\left[\frac{1}{s^2 - a^2}\right]$	$\frac{1}{a} \sinh at$	$L^{-1}\left[\frac{1}{s^2 - 9}\right] = \frac{1}{3} \sinh 3t$

Properties of Inverse Laplace Transforms :

1. Linearity Property : Let $L[f(t)] = F(s)$ and $L[g(t)] = G(s)$ and let c_1 and c_2 be any two arbitrary constants. Then,

$$L^{-1}[c_1 F(s) + c_2 G(s)] = c_1 L^{-1}[F(s)] + c_2 L^{-1}[G(s)] = c_1 f(t) + c_2 g(t)$$

Example : Find the inverse Laplace Transforms of $\frac{5s-4}{s^2+4}$

Solution : Using Linearity Property,

$$\begin{aligned} L^{-1}\left[\frac{5s-4}{s^2+4}\right] &= 5L^{-1}\left[\frac{s}{s^2+4}\right] - 4L^{-1}\left[\frac{1}{s^2+4}\right] \\ &= 5\cos 2t - 4\left(\frac{1}{2}\right)\sin 2t \\ &= \boxed{5\cos 2t - 2\sin 2t} \end{aligned}$$

2. First Shift Property : If $L^{-1}[F(s)] = f(t)$ then,

$$L^{-1}[F(s-a)] = e^{at} L^{-1}[F(s)] = e^{at} f(t)$$

Example : Find the inverse Laplace Transforms of $\frac{2s+3}{s^2+2s+2}$

$$\begin{aligned} \text{Solution : } L^{-1}\left[\frac{2s+3}{s^2+2s+2}\right] &= L^{-1}\left[\frac{2s+3}{(s+1)^2+1}\right] \\ &= L^{-1}\left[\frac{2(s+1)+1}{(s+1)^2+1}\right] \\ &= 2L^{-1}\left[\frac{s+1}{(s+1)^2+1}\right] + L^{-1}\left[\frac{1}{(s+1)^2+1}\right] \\ &= 2e^{-t}L^{-1}\left[\frac{s}{s^2+1}\right] + e^{-t}L^{-1}\left[\frac{1}{s^2+1}\right] \\ &= \boxed{2e^{-t}\cos t + e^{-t}\sin t} \end{aligned}$$

3. **Change of Scale Property** : If $L^{-1}[F(s)] = f(t)$ then,

$$L^{-1}[F(ks)] = \frac{1}{k} f\left(\frac{t}{k}\right)$$

Example : If $L^{-1}\left\{\frac{s^2-1}{(s^2+1)^2}\right\} = t \cos t$

find $L^{-1}\left\{\frac{9s^2-1}{(9s^2+1)^2}\right\}$

Solution : Let $F(s) = \frac{s^2-1}{(s^2+1)^2}$.

Then, $L^{-1}[F(s)] = L^{-1}\left[\frac{s^2-1}{(s^2+1)^2}\right] = t \cos t = f(t)$.

Therefore, $L^{-1}[F(3s)] = L^{-1}\left[\frac{(3s)^2-1}{[(3s)^2+1]^2}\right] = \frac{1}{3} \cdot \frac{t}{3} \cos \frac{t}{3} =$

$$\frac{t}{9} \cos \frac{t}{3}$$

4. Inverse Laplace Transforms – Method of Partial Fractions

Sometimes the given $F(s) = \frac{P(s)}{Q(s)}$ where $P(s)$ and $Q(s)$ are polynomials in s can be expressed as partial fractions for obtaining the Inverse Laplace Transform

Pre – Requisite :

Sl No.	Factors in the denominator	Corresponding Partial Fractions
1.	Non- repeated linear factors $F(s) = \frac{s}{(s+2)(3s+5)}$	$F(s) = \frac{A}{s+2} + \frac{B}{3s+5}, A \neq 0 \text{ \& } B \neq 0$
2.	Repeated linear factor $F(s) = \frac{2s+5}{(s+1)^3}$	$F(s) = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{(s+1)^3},$ $A \neq 0, B \neq 0 \text{ \& } C \neq 0$
3.	Non – repeated quadratic factor $F(s) = \frac{2s+1}{(s^2+2s+2)(s^2+2s+5)}$	$F(s) = \frac{As+B}{(s^2+2s+2)} + \frac{Cs+D}{(s^2+2s+5)}$

4.	<u>Repeated quadratic factor</u> $F(s) = \frac{2s + 1}{(s^2 + 2s + 5)^2}$	$F(s) = \frac{As + B}{(s^2 + 2s + 5)} + \frac{Cs + D}{(s^2 + 2s + 5)^2}$
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Example : Obtain the Inverse Laplace Transforms of $\frac{s-2}{s^2+5s+6}$

Solution : Consider, $\frac{s-2}{s^2+5s+6} = \frac{s-2}{(s+2)(s+3)} = \frac{A}{s+2} + \frac{B}{s+3}$

Here , $A = -4$, $B = 5$

Therefore, $L^{-1} \left[\frac{s-2}{s^2+5s+6} \right] = -4L^{-1} \left[\frac{1}{s+2} \right] + 5L^{-1} \left[\frac{1}{s+3} \right] = -4e^{-2t} + 5e^{-3t}$

4. Inverse Laplace Transform of Derivatives : If $L^{-1}[F(s)] = f(t)$ then,

$L^{-1}[F^{(n)}(s)] = (-1)^n t^n f(t)$, $n = 1, 2, 3, \dots$

Example : Find the inverse Laplace Transforms of $\frac{s}{(s^2+a^2)^2}$

Solution : Let $F(s) = \frac{1}{s^2+a^2}$.

Then, $L^{-1}[F(s)] = L^{-1} \left[\frac{1}{s^2+a^2} \right] = \frac{\sin at}{a} = f(t)$.

and $F'(s) = \frac{d}{ds} \left\{ \frac{1}{s^2+a^2} \right\} = \frac{-2s}{(s^2+a^2)^2}$

Taking inverse, $L^{-1} \left[\frac{-2s}{(s^2+a^2)^2} \right] = L^{-1}[F'(s)] = (-1)^1 t f(t)$

Therefore, $L^{-1} \left[\frac{s}{(s^2+a^2)^2} \right] = t f(t) = \frac{t \sin at}{2a}$

5. Inverse Laplace Transform of Integrals : If $L^{-1}[F(s)] = f(t)$ then,

$$L^{-1} \left[\int_s^\infty F(s) ds \right] = \frac{f(t)}{t}$$

Example : Obtain the inverse Laplace Transforms of

$$L^{-1} \left[\int_s^\infty \left(\frac{1}{s} - \frac{1}{s+1} \right) ds \right]$$

Solution : Let $F(s) = \frac{1}{s} - \frac{1}{s+1}$

$$\text{Then } L^{-1} \left[\int_s^\infty \left(\frac{1}{s} - \frac{1}{s+1} \right) ds \right] = \frac{L^{-1} \left[\frac{1}{s} - \frac{1}{s+1} \right]}{t} = \frac{1 - e^{-t}}{t}$$

6. Inverse Laplace Transform when ' $F(s)$ ' is multiplied by ' s ' : If $L^{-1}[F(s)] = f(t)$ and $f(0) = 0$ then, $L^{-1}[sF(s)] = \frac{df(t)}{dt}$

Example : Obtain the inverse Laplace Transform of $\frac{8s}{s^2+9}$

Solution : Let $F(s) = \frac{8}{s^2+9}$.

$$\text{Then } L^{-1}[F(s)] = L^{-1} \left[\frac{8}{s^2+9} \right] = 8L^{-1} \left[\frac{1}{s^2+9} \right] = \frac{8}{3} \sin 3t = f(t)$$

Therefore,

$$L^{-1}[sF(s)] = L^{-1} \left[\frac{8s}{s^2+9} \right] = \frac{df(t)}{dt} = \frac{d}{dt} \left\{ \frac{8}{3} \sin 3t \right\} = 8 \cos 3t$$

7. Inverse Laplace Transform when ' $F(s)$ ' is divided by powers of ' s ' : If $L^{-1}[F(s)] = f(t)$ then,

$$L^{-1} \frac{F(s)}{s} = \int_0^t f(t) dt$$

$$\text{Note : } L^{-1} \frac{F(s)}{s^n} = \int_0^t \int_0^t \dots \int_0^t f(t) dt^n$$

Example : Find the inverse Laplace transform of $\frac{1}{s(s+4)}$

Solution : Let $F(s) = \frac{1}{s(s+4)}$.

$$\text{Then } L^{-1}[F(s)] = L^{-1} \left[\frac{1}{s(s+4)} \right] = e^{-4t} = f(t).$$

$$\text{Therefore, } L^{-1} \left[\frac{1}{s(s+4)} \right] = \int_0^t e^{-4t} dt = \left[\frac{e^{-4t}}{-4} \right]_0^t = \frac{1 - e^{-4t}}{4}$$

8. Second Shifting theorem : If $L^{-1}[F(s)] = f(t)$ then,

$$L^{-1}[e^{-as}F(s)] = f(t - a)u(t - a)$$

Example : Find the inverse Laplace Transform of $\frac{e^{-3s}}{s^2+4}$

Solution : Let $F(s) = \frac{1}{s^2+4}$

$$L^{-1}[F(s)] = \frac{1}{2} \sin 2t = f(t)$$

$$\begin{aligned} L^{-1}\left[\frac{e^{-3s}}{s^2 + 4}\right] &= f(t-3) u(t-3) \\ &= \frac{1}{2} \sin 2(t-3) u(t-3) \end{aligned}$$

Convolution

Definition : The convolution of two functions $f(t)$ and $g(t)$ is denoted by $(f * g)(t)$ and is defined as

$$(f * g)(t) = \int_0^t f(u)g(t-u)du$$

is called as the convolution or faltung of ' f ' and ' g ' and is regarded as a generalized product of these functions.

Convolution theorem : If $L^{-1}[F(s)] = f(t)$ and $L^{-1}[G(s)] = g(t)$ then,

$$L^{-1}[F(s).G(s)] = f(t) * g(t)$$

Application of Laplace Transform to Differential Equations with constant co-efficients

The Laplace Transforms are used to solve differential equations since it yields the particular solutions without the necessity of first finding the general solution and then evaluating the arbitrary constants.

Working Procedure :

Step 1. Take Laplace Transforms on both sides of the given Differential equation.

Step 2. Solve the algebraic equation to get y in terms of s .

Step 3. Resolve the function of s into partial fractions.

Step 4. Take the inverse Laplace transform on both sides to obtain the desired solution satisfying the given conditions.

Note : Instead of Step 3 , convolution theorem can be used.

Problems :

Obtain the inverse Laplace transform of the following :

1. $\frac{3s-12}{s^2+8}$

Solution:

$$\text{Let } F(s) = \frac{3s-12}{s^2+8} = \frac{3s}{s^2+8} - \frac{12}{s^2+8}$$

$$\Rightarrow F(s) = 3L^{-1}\left(\frac{s}{s^2+8}\right) - 12L^{-1}\left(\frac{1}{s^2+8}\right)$$

$$\Rightarrow F(s) = 3L^{-1}\left(\frac{s}{s^2+(2\sqrt{2})^2}\right) - 12L^{-1}\left(\frac{1}{s^2+(2\sqrt{2})^2}\right)$$

$$\Rightarrow F(s) = 3\cos 2\sqrt{2}t - 3\sqrt{2}\sin 2\sqrt{2}t.$$

2. $\left(\frac{\sqrt{s}-1}{s}\right)^2$

Solution:

$$\text{Let } F(s) = \left(\frac{\sqrt{s}-1}{s}\right)^2$$

$$\Rightarrow F(s) = \left(\frac{s+1-2\sqrt{s}}{s^2}\right)$$

$$\Rightarrow F(s) = \left(\frac{1}{s} + \frac{1}{s^2} - \frac{2}{s^{\frac{3}{2}}}\right), \text{ since } L^{-1}\left(\frac{1}{s^{n+1}}\right) = \frac{t^n}{\Gamma(n+1)}$$

$$\Rightarrow f(t) = L^{-1}(F(s)) = 1 + t - 2\left(\frac{2\sqrt{t}}{\sqrt{\pi}}\right) = 1 + t - 4\left(\frac{\sqrt{t}}{\sqrt{\pi}}\right)$$

3. $\frac{1}{s} \sin\left(\frac{1}{s}\right)$

Solution :

$$\text{Let } F(s) = \frac{1}{s} \sin\left(\frac{1}{s}\right)$$

$$\text{We have } \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\Rightarrow \sin\left(\frac{1}{s}\right) = \frac{1}{s} - \frac{1}{3!}\left(\frac{1}{s}\right)^3 + \frac{1}{5!}\left(\frac{1}{s}\right)^5 - \frac{1}{7!}\left(\frac{1}{s}\right)^7 + \dots$$

$$\Rightarrow \frac{1}{s} \cdot \sin\left(\frac{1}{s}\right) = \frac{1}{s^2} - \frac{1}{3!}\left(\frac{1}{s}\right)^4 + \frac{1}{5!}\left(\frac{1}{s}\right)^6 - \frac{1}{7!}\left(\frac{1}{s}\right)^8 + \dots$$

$$\Rightarrow f(t) = L^{-1}(F(s)) = t - \frac{1}{3!} \cdot \frac{t^3}{3!} + \frac{1}{5!} \cdot \frac{t^5}{5!} - \frac{1}{7!} \cdot \frac{t^7}{7!} + \dots$$

$$\Rightarrow f(t) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} t^{2n-1}}{(2n-1)!^2}$$

4. $\frac{s}{(s+1)^5}$

Solution :

Let $F(s) = \frac{s}{(s+1)^5}$

$$\Rightarrow F(s) = \frac{(s+1)-1}{(s+1)^5} = \frac{1}{(s+1)^4} - \frac{1}{(s+1)^5}$$

$$\Rightarrow f(t) = L^{-1}(F(s)) = e^{-t}L^{-1}\left(\frac{1}{s^4} - \frac{1}{s^5}\right)$$

$$\Rightarrow f(t) = \frac{e^{-t}}{24}(4t^3 - t^4)$$

5. $\frac{5s-2}{s^2+4s+8}$

Solution :

Let $F(s) = \frac{5s-2}{s^2+4s+8} = \frac{5(s-2+2)-2}{(s+2)^2+4} = \frac{5(s+2)-12}{(s+2)^2+4}$

$$\Rightarrow f(t) = L^{-1}(F(s)) = 5e^{-2t}L^{-1}\left(\frac{s}{s^2+4}\right) - \frac{12}{2}e^{-2t}L^{-1}\left(\frac{2}{s^2+4}\right)$$

$$\Rightarrow f(t) = e^{-2t}(5\cos 2t - 6\sin 2t)$$

6. $\tan^{-1} \frac{2}{s^2}$

Solution :

Let $F(s) = \tan^{-1} \frac{2}{s^2}$

We have $L(f(t)) = F(s) = \tan^{-1} \frac{2}{s^2}$

$$\Rightarrow L(tf(t)) = -\frac{d}{ds}\left(\tan^{-1} \frac{2}{s^2}\right)$$

$$\Rightarrow L(tf(t)) = \frac{4s}{s^4+4}$$

$$\Rightarrow tf(t) = L^{-1}\left(\frac{4s}{s^4+4}\right) = L^{-1}\frac{4s}{(s^2+2s+2)(s^2-2s+2)}$$

Now, using partial fractions, we obtain, $\frac{4s}{(s^2+2s+2)(s^2-2s+2)} = \frac{1}{s^2-2s+2} - \frac{1}{s^2+2s+2}$

$$\Rightarrow L^{-1}\left(\frac{4s}{s^4+4}\right) = L^{-1}\left(\frac{1}{s^2-2s+2}\right) - L^{-1}\left(\frac{1}{s^2+2s+2}\right)$$

$$\begin{aligned}\Rightarrow L^{-1}\left(\frac{4s}{s^4+4}\right) &= L^{-1}\left(\frac{1}{(s-1)^2+1}\right) - L^{-1}\left(\frac{1}{(s+1)^2+1}\right) \\ \Rightarrow L^{-1}\left(\frac{4s}{s^4+4}\right) &= e^t L^{-1}\left(\frac{1}{s^2+1}\right) - e^{-t} L^{-1}\left(\frac{1}{s^2+1}\right) \\ \Rightarrow L^{-1}\left(\frac{4s}{s^4+4}\right) &= e^t \sin t - e^{-t} \sin t = 2 \sin t \sinh t \\ \Rightarrow f(t) &= \frac{2 \sin t \sinh t}{t}\end{aligned}$$

7. $\log\left(1 - \frac{a^2}{s^2}\right)$

Solution :

$$\begin{aligned}\text{Let } F(s) &= \log\left(1 - \frac{a^2}{s^2}\right) = \log(s^2 - a^2) - 2 \log s \\ \text{We have } L[f(t)] &= F(s) \\ \Rightarrow L[f(t)] &= \log(s^2 - a^2) - 2 \log s \\ \Rightarrow L[tf(t)] &= -\frac{d}{ds} [\log(s^2 - a^2) - 2 \log s] \\ \Rightarrow L[tf(t)] &= \left[\frac{2}{s} - \frac{2s}{s^2 - a^2} \right] \\ \Rightarrow tf(t) &= L^{-1}\left(\frac{2}{s} - \frac{2s}{s^2 - a^2}\right) = 2 - 2 \cosh at \\ \Rightarrow f(t) &= \frac{2(1 - \cosh at)}{t}\end{aligned}$$

8. $\cot^{-1}\left(\frac{s+3}{2}\right)$

Solution :

$$\begin{aligned}\text{Let } F(s) &= \cot^{-1}\left(\frac{s+3}{2}\right) \\ \text{We have } L[f(t)] &= F(s) \\ \Rightarrow L[f(t)] &= \cot^{-1}\left(\frac{s+3}{2}\right) \\ \Rightarrow L[tf(t)] &= -\frac{d}{ds} \cot^{-1}\left(\frac{s+3}{2}\right) = \frac{2}{(s+3)^2 + 4} \\ \Rightarrow tf(t) &= L^{-1}\left(\frac{2}{(s+3)^2 + 4}\right)\end{aligned}$$

$$\Rightarrow tf(t) = e^{-3t} L^{-1} \left(\frac{2}{s^2 + 4} \right) = e^{-3t} \sin 2t$$

$$\Rightarrow f(t) = \frac{e^{-3t} \sin 2t}{t}$$

9. $\frac{s}{(s^2+1)^2}$

Solution :

Let $F(s) = \frac{1}{(s^2+1)}$, so that $f(t) = L^{-1}(F(s)) = \sin t$.

Then $F'(s) = \frac{-2s}{(s^2+1)^2}$. By inverse Laplace transform of derivatives with $s = 1$,

we get $L^{-1}(F'(s)) = -tf(t)$

$$\Rightarrow L^{-1} \left(\frac{s}{(s^2+1)^2} \right) = \frac{t \sin t}{2}$$

10. $\frac{1}{s^2} \left(\frac{s+1}{s^2+1} \right)$

Solution :

$$\text{Let } F(s) = \frac{1}{s^2} \cdot \frac{s+1}{s^2+1} = \frac{1}{s} \left(\frac{1}{s} \left(\frac{s+1}{s^2+1} \right) \right)$$

$$\text{Now, } L^{-1} \left[\frac{1}{s} \cdot \frac{s+1}{s^2+1} \right] = \int_0^t (\cos t + \sin t) dt$$

$$\Rightarrow L^{-1} \left[\frac{1}{s} \cdot \frac{s+1}{s^2+1} \right] = (\sin t - \cos t)_0^t = \sin t - \cos t + 1$$

$$\Rightarrow L^{-1} \left[\frac{1}{s^2} \left(\frac{s+1}{s^2+1} \right) \right] = \int_0^t (\sin t - \cos t + 1) dt$$

$$\Rightarrow L^{-1} \left[\frac{1}{s^2} \left(\frac{s+1}{s^2+1} \right) \right] = (-\cos t - \sin t + t)_0^t$$

$$\Rightarrow L^{-1} \left[\frac{1}{s^2} \left(\frac{s+1}{s^2+1} \right) \right] = 1 + t - \cos t - \sin t$$

11. $\frac{1}{s(s^2+a^2)}$

Solution :

$$\text{Let } F(s) = \frac{1}{s(s^2+a^2)}$$

$$\text{Now, } L^{-1} \left(\frac{1}{s(s^2+a^2)} \right) = \frac{\sin at}{a}$$

$$\Rightarrow L^{-1}\left(\frac{1}{s} \cdot \frac{1}{s^2+a^2}\right) = \int_0^t \left(\frac{\sin at}{a}\right) dt = \left(\frac{-\cos at}{a^2}\right)_0^t = \frac{1-\cos at}{a^2}$$

12. $\frac{e^{-4s}-e^{-7s}}{s^2}$

Solution :

Let $F(s) = \frac{1}{s^2}$

$$\Rightarrow L^{-1}(F(s)) = t$$

By second shifting theorem, we have $L^{-1}(F(s)) = (t-4)u(t-4) - (t-7)u(t-7)$

Also,

$$u(t-4) = \begin{cases} 0, & 0 < t < 4 \\ 1, & t > 4 \end{cases}$$

and

$$u(t-7) = \begin{cases} 0, & 0 < t < 7 \\ 1, & t > 7 \end{cases}$$

$$\Rightarrow L^{-1}\left[\frac{e^{-4s}-e^{-7s}}{s^2}\right] = \begin{cases} 0-0, & \text{if } 0 < t < 4 \\ (t-4)-0, & \text{if } 4 < t < 7 \\ (t-4)-(t-7), & \text{if } t > 7 \end{cases}$$

$$\Rightarrow L^{-1}\left[\frac{e^{-4s}-e^{-7s}}{s^2}\right] = \begin{cases} 0, & \text{if } 0 < t < 4 \\ (t-4), & \text{if } 4 < t < 7 \\ 3, & \text{if } t > 7 \end{cases}$$

13. $\frac{3}{s} - 4\left(\frac{e^{-s}}{s^2}\right) + 4\left(\frac{e^{-3s}}{s^2}\right)$

Solution :

Let $F(s) = \frac{3}{s} - 4\left(\frac{1}{s^2}\right) + 4\left(\frac{1}{s^2}\right)$

Since $L^{-1}\left(\frac{1}{s}\right) = 1$ and $L^{-1}\left(\frac{1}{s^2}\right) = t$,

using second shifting theorem, we have

$$L^{-1}\left[\frac{3}{s} - 4\left(\frac{e^{-s}}{s^2}\right) + 4\left(\frac{e^{-3s}}{s^2}\right)\right] = 3 - 4(t-1)u(t-1) + 4(t-3)u(t-3).$$

14. $\frac{s^2+9s-9}{s^3-9s}$

Solution :

$$\text{Let } F(s) = \frac{s^2+9s-9}{s^3-9s} = \frac{s^2+9s-9}{s(s^2-9)} = \frac{s^2+9s-9}{s(s+3)(s-3)}$$

$$\text{Using partial fractions, } \frac{s^2+9s-9}{s(s+3)(s-3)} = \frac{A}{s} + \frac{B}{s+3} + \frac{C}{s-3}$$

$$\Rightarrow s^2 + 9s - 9 = A(s+3)(s-3) + BS(s-3) + Cs(s+3)$$

For $s = 0, s = 3, s = -3$, we find $A = 1, C = \frac{3}{2}, B = \frac{-3}{2}$, respectively.

$$\Rightarrow F(s) = \frac{s^2 + 9s - 9}{s^3 - 9s} = \frac{s^2 + 9s - 9}{s(s^2 - 9)} = \frac{1}{s} + \frac{\frac{-3}{2}}{s+3} + \frac{\frac{3}{2}}{s-3}$$

$$\Rightarrow L^{-1}(F(s)) = 1 - \frac{3}{2}e^{-3t} + \frac{3}{2}e^{3t} = 1 + 3\left(\frac{e^{3t} + e^{-3t}}{2}\right) = 1 + 3\sinh 3t$$

15. $\frac{s+1}{(s^2+1)(s^2+4)}$

Solution :

$$\text{Let } F(s) = \frac{s+1}{(s^2+1)(s^2+4)}$$

$$\text{Using partial fractions, } \frac{s+1}{(s^2+1)(s^2+4)} = \frac{As+B}{s^2+1} + \frac{Cs+D}{s^2+4}$$

$$\Rightarrow s + 1 = (As + B)(s^2 + 4) + (Cs + D)(s^2 + 1)$$

On comparing the coefficients of:

$$s^3: 0 = A + C$$

$$s^2: 0 = B + D$$

$$s: 1 = 4A + C$$

$$\text{constant: } 1 = 4B + D.$$

On solving first and third equations, we get $A = \frac{1}{3}; C = \frac{-1}{3}$

Similarly, on solving second and fourth equations, we get $B = \frac{1}{3}; D = \frac{-1}{3}$

$$\Rightarrow F(s) = \frac{\frac{1}{3}s + \frac{1}{3}}{s^2 + 1} + \frac{\frac{-1}{3}s + \frac{-1}{3}}{s^2 + 4}$$

$$\Rightarrow F(s) = \frac{1}{3}\left(\frac{s}{s^2 + 1} + \frac{1}{s^2 + 1}\right) - \frac{1}{3}\left(\frac{s}{s^2 + 4} + \frac{1}{s^2 + 4}\right)$$

$$\Rightarrow f(t) = L^{-1}(F(s)) = \frac{1}{3}(\cos t + \sin t) - \frac{1}{3}\left(\cos 2t + \frac{\sin 2t}{2}\right)$$

$$\Rightarrow f(t) = \frac{1}{6}(2\cos t - 2\cos 2t + 2\sin t - \sin 2t)$$

16.
$$\frac{5s^2-7s+17}{(s-1)(s^2+4)}$$

Solution :

Let
$$F(s) = \frac{5s^2-7s+17}{(s-1)(s^2+4)}$$

Using partial fractions,
$$\frac{5s^2-7s+17}{(s-1)(s^2+4)} = \frac{A}{s-1} + \frac{Bs+C}{s^2+4}$$

$$\Rightarrow 5s^2 - 7s + 17 = A(s^2 + 4) + (Bs + C)(s - 1)$$

Putting $s = 1$, we get $5A = 15 \Rightarrow A = 3$

Putting $s = 0$, we get $A + B = 5 \Rightarrow B = 2$

Comparing the coefficients of s^2 , we get $4A - C = 17 \Rightarrow C = -5$

$$\Rightarrow F(s) = \frac{3}{s-1} + \frac{2s-5}{s^2+4} = \frac{3}{s-1} + \frac{2s}{s^2+4} - \frac{5}{s^2+4}$$

$$\Rightarrow f(t) = L^{-1}(F(s)) = 3e^t + 2\cos 2t - \frac{5}{2}\sin 2t$$

17.
$$\int_s^\infty \log\left(\frac{s+2}{s+1}\right) ds$$

Solution :

Let
$$F(s) = \log\left(\frac{s+2}{s+1}\right) = \log(s+2) - \log(s+1)$$

Now,
$$L(f(t)) = F(s)$$

$$\Rightarrow L(f(t)) = \log(s+2) - \log(s+1)$$

$$\Rightarrow L(tf(t)) = -\frac{d}{ds}(F(s)) = -\frac{d}{ds}(\log(s+2) - \log(s+1))$$

$$\Rightarrow L(tf(t)) = -\left(\frac{1}{s+2} - \frac{1}{s+1}\right)$$

$$\Rightarrow tf(t) = -L^{-1}\left(\frac{1}{s+2} - \frac{1}{s+1}\right)$$

$$\Rightarrow tf(t) = e^{-t} - e^{-2t}$$

$$\Rightarrow f(t) = \frac{e^{-t} - e^{-2t}}{t}$$

Therefore,
$$L^{-1} \int_s^\infty \log\left(\frac{s+2}{s+1}\right) ds = \frac{e^{-t} - e^{-2t}}{t^2}$$

18. Using Convolution theorem to obtain the inverse Laplace transform of the following:

a. $\frac{1}{(s+1)(s^2+1)}$

Solution :

Let $F(s) = \frac{1}{(s+1)(s^2+1)}$

Now, $L^{-1}\left(\frac{1}{(s+1)(s^2+1)}\right) = L^{-1}\left(\frac{1}{s+1} \cdot \frac{1}{s^2+1}\right) = e^{-t} * \sin t = \sin t * e^{-t}$

$\Rightarrow L^{-1}\left(\frac{1}{(s+1)(s^2+1)}\right) = \int_0^t \sin u \cdot e^{-(t-u)} du$

$\Rightarrow L^{-1}\left(\frac{1}{(s+1)(s^2+1)}\right) = e^{-t} \int_0^t e^u \cdot \sin u du$

$\Rightarrow L^{-1}\left(\frac{1}{(s+1)(s^2+1)}\right) = e^{-t} \left(\frac{e^u}{2} (\sin u - \cos u) \right)_0^t$

$\Rightarrow L^{-1}\left(\frac{1}{(s+1)(s^2+1)}\right) = \frac{e^{-t}}{2} (e^t (\sin t - \cos t) + 1)$

$\Rightarrow L^{-1}\left(\frac{1}{(s+1)(s^2+1)}\right) = \frac{1}{2} (\sin t - \cos t + e^{-t}).$

b. $\frac{1}{(s^2+4)(s+1)^2}$

Solution :

Let $F(s) = \frac{1}{(s^2+4)(s+1)^2} = \frac{1}{(s^2+4)(s+1)(s+1)} = F_1(s) \cdot F_2(s) \cdot F_3(s)$, say.

$\Rightarrow L^{-1}(F_1(s)) = \frac{\sin 2t}{2}; L^{-1}(F_2(s)) = e^{-t}; L^{-1}(F_3(s)) = e^{-t}.$

By Convolution theorem, $L^{-1}(F_1(s) \cdot F_2(s)) = \int_0^t \frac{\sin 2u}{2} e^{-(t-u)} du$

$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s)) = \frac{e^{-t}}{2} \int_0^t e^u \sin 2u du$

$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s)) = \frac{e^{-t}}{2} \frac{e^u}{5} (\sin 2u - 2 \cos 2u)_0^t$

$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s)) = \frac{e^{-t}}{10} e^t (\sin 2t - 2 \cos 2t) + 2$

$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s)) = \frac{\sin 2t - 2 \cos 2t}{10} + \frac{e^{-t}}{5}$

$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s) \cdot F_3(s)) = \int_0^t \left(\frac{\sin 2u - 2 \cos 2u}{10} + \frac{e^{-u}}{5} \right) (e^{-(t-u)}) du$

$$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s) \cdot F_3(s)) = e^{-t} \int_0^t \left(\frac{\sin 2u - 2\cos 2u}{10} + \frac{e^{-u}}{5} \right) (e^u) du$$

$$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s) \cdot F_3(s)) = e^{-t} \int_0^t \left(\frac{e^u(\sin 2u - 2\cos 2u)}{10} + \frac{1}{5} \right) du$$

$$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s) \cdot F_3(s)) = e^{-t} \int_0^t \left(\frac{e^u(\sin 2u - 2\cos 2u) + 2}{10} \right) du$$

$$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s) \cdot F_3(s)) = \frac{e^{-t}}{10} \int_0^t (e^u(\sin 2u - 2\cos 2u) + 2) du$$

$$\text{Now, } \int_0^t e^u \sin 2u du = \frac{e^u}{5} (\sin 2u - 2\cos 2u) \Big|_0^t$$

$$\Rightarrow \int_0^t e^u \sin 2u du = \frac{e^t}{5} (\sin 2t - 2\cos 2t) + \frac{2}{5}$$

$$\text{Similarly, } \int_0^t e^u \cos 2u du = \frac{e^u}{5} (\cos 2u + 2\sin 2u) \Big|_0^t$$

$$\Rightarrow \int_0^t e^u \cos 2u du = \frac{e^t}{5} (\cos 2t + 2\sin 2t) - \frac{1}{5}$$

$$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s) \cdot F_3(s)) = \frac{e^{-t}}{10} \frac{e^t}{5} (\sin 2t - 2\cos 2t) + \frac{2}{5} - \frac{2e^t}{5} (\cos 2t + 2\sin 2t) + \frac{2}{5} + 2t$$

$$\Rightarrow L^{-1}(F_1(s) \cdot F_2(s) \cdot F_3(s)) = \frac{2}{25} e^{-t} + \frac{te^{-t}}{5} - \frac{1}{50} (3\sin 2t + 4\cos 2t)$$

Using Laplace transforms solve the following differential equations consisting of initial conditions :

$$1. y' - y = e^{3t}, y(0) = 2.$$

Solution :

$$\text{Given } y' - y = e^{3t}.$$

Taking Laplace transform on both sides, we get

$$L(y') - L(y) = L(e^{3t})$$

$$\Rightarrow sL(y(t)) - y(0) - L(y(t)) = \frac{1}{s-3}$$

$$\Rightarrow sL(y(t)) - 2 - L(y(t)) = \frac{1}{s-3} (\text{Using initial conditions})$$

$$\Rightarrow (s-1)L(y(t)) = \frac{1}{s-3} + 2$$

$$\Rightarrow (s-1)L(y(t)) = \frac{2s-5}{s-3}$$

$$\Rightarrow y(t) = L^{-1} \left(\frac{2s-5}{(s-3)(s-1)} \right)$$

Using partial fractions, $\frac{2s-5}{(s-3)(s-1)} = \frac{A}{s-3} + \frac{B}{s-1}$

$$\Rightarrow 2s - 5 = A(s - 1) + B(s - 3).$$

Here, $A = \frac{1}{2}; B = \frac{3}{2}$

$$\Rightarrow y(t) = L^{-1} \left(\frac{\frac{1}{2}}{s-3} + \frac{\frac{3}{2}}{s-1} \right) = \frac{1}{2}e^{3t} + \frac{3}{2}e^t$$

2. $y'' - 6y' + 9y = 0, y(0) = 2; y'(0) = 9.$

Solution :

Given $y'' - 6y' + 9y = 0.$

Taking Laplace transform on both sides, we get

$$L(y'') - 6L(y') + 9L(y) = 0$$

$$\Rightarrow (s^2L(y(t)) - sy(0) - y'(0)) - 6(sL(y(t)) - y(0)) + 9L(y(t)) = 0$$

$$\Rightarrow s^2L(y(t)) - 2s - 9 - 6(sL(y(t)) - 2) + 9L(y(t)) = 0 \text{ (Using initial conditions)}$$

$$\Rightarrow (s^2 - 6s + 9)L(y(t)) = 2s + 9 - 12$$

$$\Rightarrow L(y(t)) = \frac{2s - 3}{s^2 - 6s + 9} = \frac{2s - 3}{(s - 3)^2}$$

$$\Rightarrow y(t) = L^{-1} \left(\frac{2s-3}{(s-3)^2} \right)$$

$$\Rightarrow y(t) = L^{-1} \left(\frac{2(s-3) + 3}{(s-3)^2} \right)$$

$$\Rightarrow y(t) = e^{3t}L^{-1} \left(\frac{2s+3}{s^2} \right)$$

$$\Rightarrow y(t) = e^{3t}L^{-1} \left(\frac{2}{s} \right) + L^{-1} \left(\frac{3}{s^2} \right)$$

$$\Rightarrow y(t) = e^{3t}2 + 3t$$

3. $y'''' - 16y = 30\sin t, y''(0) = 0; y''(\pi) = 0, y'''(0) = -18; y'''(\pi) = -18.$

Solution :

Given $y'''' - 16y = 30\sin t.$

Taking Laplace transform on both sides, we get

$$L(y''''') - 16L(y) = 30L(\sin t)$$

$$\Rightarrow (s^4 L[y(t)] - s^3 y(0) - s^2 y'(0) - s y''(0) - y'''(0)) - 16L[y(t)] = \frac{30}{s^2 + 1}$$

$$\Rightarrow (s^4 - 16)L[y(t)] = \frac{30}{s^2 + 1} + 2s^2 - 18 \text{ (Using initial conditions)}$$

$$\Rightarrow (s^4 - 16)L[y(t)] = \frac{30 + 2s^2(s^2 + 1) - 18(s^2 + 1)}{s^2 + 1}$$

$$\Rightarrow (s^4 - 16)L[y(t)] = \frac{2s^4 - 16s^2 + 12}{s^2 + 1}$$

$$\Rightarrow L[y(t)] = \frac{2s^4 - 16s^2 + 12}{(s^2 + 1)(s^4 - 16)}$$

$$\Rightarrow L[y(t)] = \frac{2s^4 - 16s^2 + 12}{(s^2 + 1)(s^2 - 4)(s^2 + 4)}$$

$$\Rightarrow y(t) = L^{-1} \left(\frac{2s^4 - 16s^2 + 12}{(s^2 + 1)(s^2 - 4)(s^2 + 4)} \right)$$

Now,

$$\frac{2s^4 - 16s^2 + 12}{(s^2 + 1)(s^2 - 4)(s^2 + 4)} = \frac{A}{s - 2} + \frac{B}{s + 2} + \frac{Cs + D}{s^2 + 1} + \frac{Es + F}{s^2 + 4}$$

$$\Rightarrow 2s^4 - 16s^2 + 12 = A(s + 2)(s^4 + 5s^2 + 4) + B(s - 2)(s^4 + 5s^2 + 4) + (Cs + D)(s^4 - 16) + (Es + F)(s^4 - 3s^2 - 4)$$

$$\text{Putting } s = 2, A = \frac{-1}{8}$$

$$\text{Putting } s = -2, B = \frac{1}{8}$$

Comparing the coefficients of :

$$s^5: A + B + C + E = 0 \Rightarrow C + E = 0$$

$$s^3: 5A + 5B - 3E = 0 \Rightarrow E = 0, \Rightarrow C = 0$$

$$s^2: 10 - 10B - 3F = -16 \Rightarrow F = \frac{-9}{2}$$

$$s^4: 2A - 2B + D + F = 2 \Rightarrow D = 7$$

$$\Rightarrow y(t) = L^{-1} \left(\frac{\frac{-1}{8}}{s - 2} + \frac{\frac{1}{8}}{s + 2} + \frac{7}{s^2 + 1} + \frac{\frac{-9}{2}}{s^2 + 4} \right)$$

$$\Rightarrow y(t) = \frac{-1}{8} e^{2t} + \frac{1}{8} e^{-2t} + 7 \sin t - \frac{9}{2} \sin 2t$$

4. An impulse voltage $E\delta(t)$ is applied to a circuit consisting of L, R, C in series with zero initial conditions. Find the limit of I as $t \rightarrow 0$ where I is the current at any subsequent time t . Equation of the circuit is $L \frac{dI}{dt} + RI + \frac{1}{C} \int_0^t I dt = E\delta(t)$, where $I(0) = 0$.

Solution :

Equation of RLC - circuit governing the circuit is

$$L \frac{dI}{dt} + RI + \frac{1}{C} \int_0^t I dt = E\delta(t), \text{ where } I(0) = 0.$$

Taking Laplace transform on both sides of the equation, we get,

$$L[I'(t)] + RL[I(t)] + \frac{1}{C} L \left[\int_0^t I dt \right] = L[E\delta(t)]$$

$$\Rightarrow [sL(I(t)) - I(0)] + RL[I(t)] + \frac{1}{C} \cdot \frac{1}{s} L[I(t)] = E, \text{ since } L[\delta(t)] = 1$$

$$\Rightarrow sL[I(t)] + RL[I(t)] + \frac{1}{C} \cdot \frac{1}{s} L[I(t)] = E, \text{ since } I(0) = 0$$

$$\Rightarrow \left(Ls + R + \frac{1}{Cs} \right) L(I(t)) = E$$

$$\Rightarrow \left(\frac{Lcs^2 + Rcs + 1}{cs} \right) L(I(t)) = E$$

$$\Rightarrow I(t) = L^{-1} \left(\frac{Ecs}{Lcs^2 + Rcs + 1} \right)$$

$$\Rightarrow I(t) = \frac{Ec}{Lc} L^{-1} \left(\frac{s}{s^2 + \frac{Rcs}{Lc} + \frac{1}{Lc}} \right)$$

$$\Rightarrow I(t) = \frac{E}{L} L^{-1} \left(\frac{s}{\left(s + \frac{R}{2L} \right)^2 + \frac{1}{Lc} - \frac{R^2}{4L^2}} \right)$$

$$\Rightarrow I(t) = \frac{E}{L} L^{-1} \left(\frac{s}{\left(s + \frac{R}{2L} \right)^2 + k^2} \right), \text{ where } k^2 = \frac{1}{Lc} - \frac{R^2}{4L^2}$$

$$\Rightarrow I(t) = \frac{E}{L} e^{-\frac{Rt}{2L}} L^{-1} \left(s - \frac{\frac{R}{2L}}{s^2 + k^2} \right)$$

$$\Rightarrow I(t) = \frac{E}{L} e^{-\frac{Rt}{2L}} \cos kt - \frac{R}{2L} \left(\frac{\sin kt}{k} \right)$$

As $t \rightarrow 0$, we have $I = \frac{E}{L}$.

Reference Books :

- i. Higher Engineering Mathematics by B S Grewal
- ii. Higher Engineering Mathematics by B V Ramana
